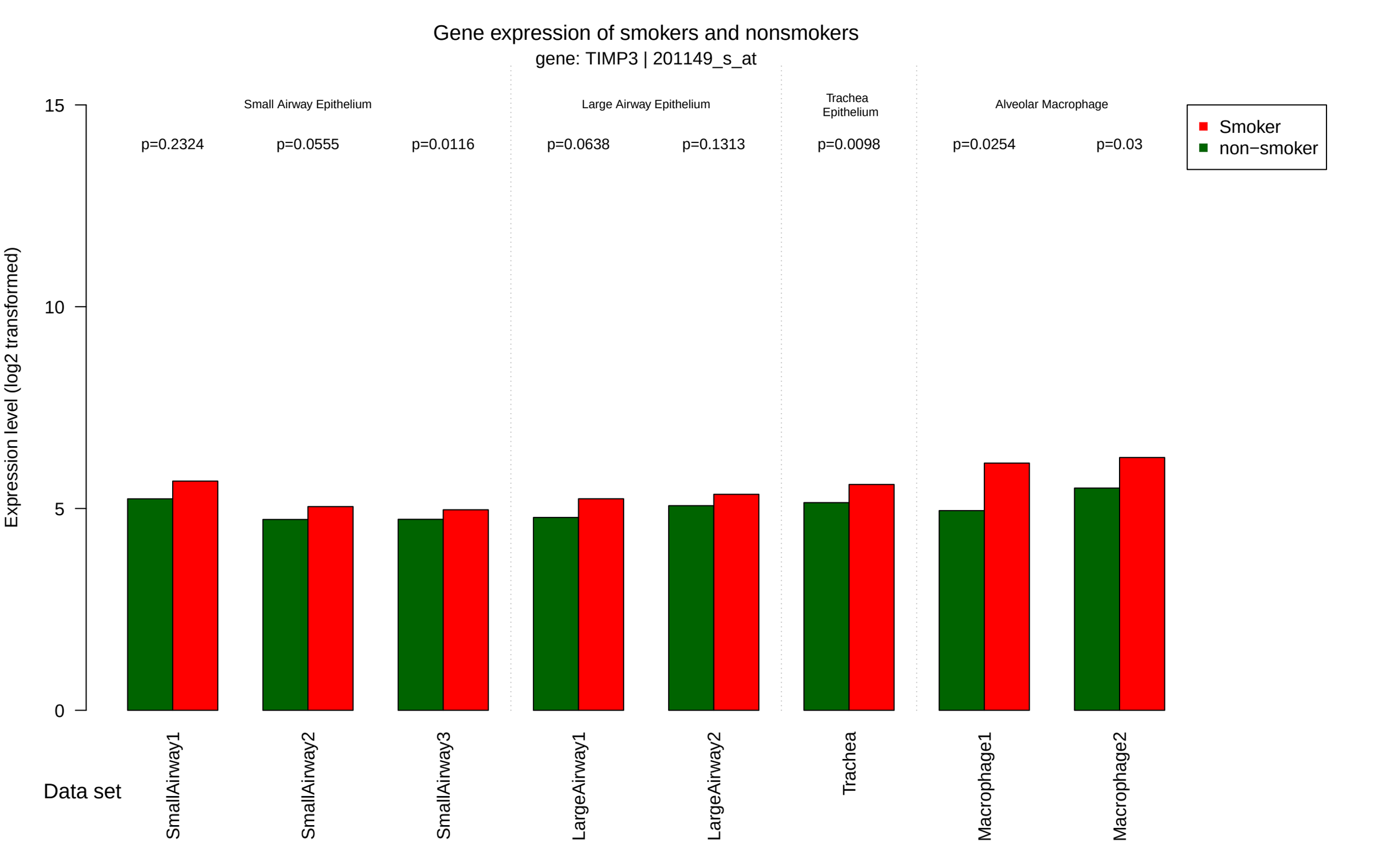
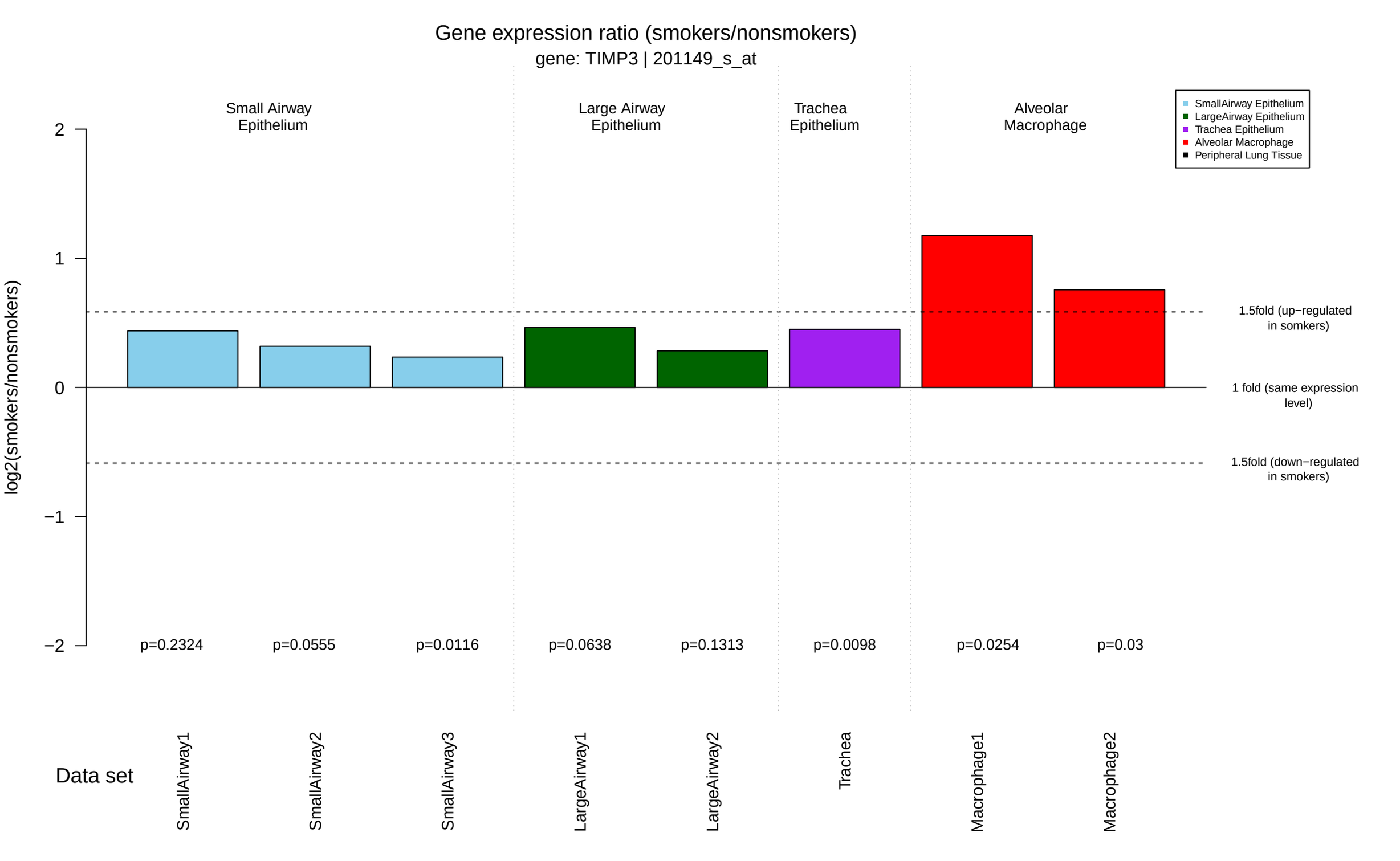
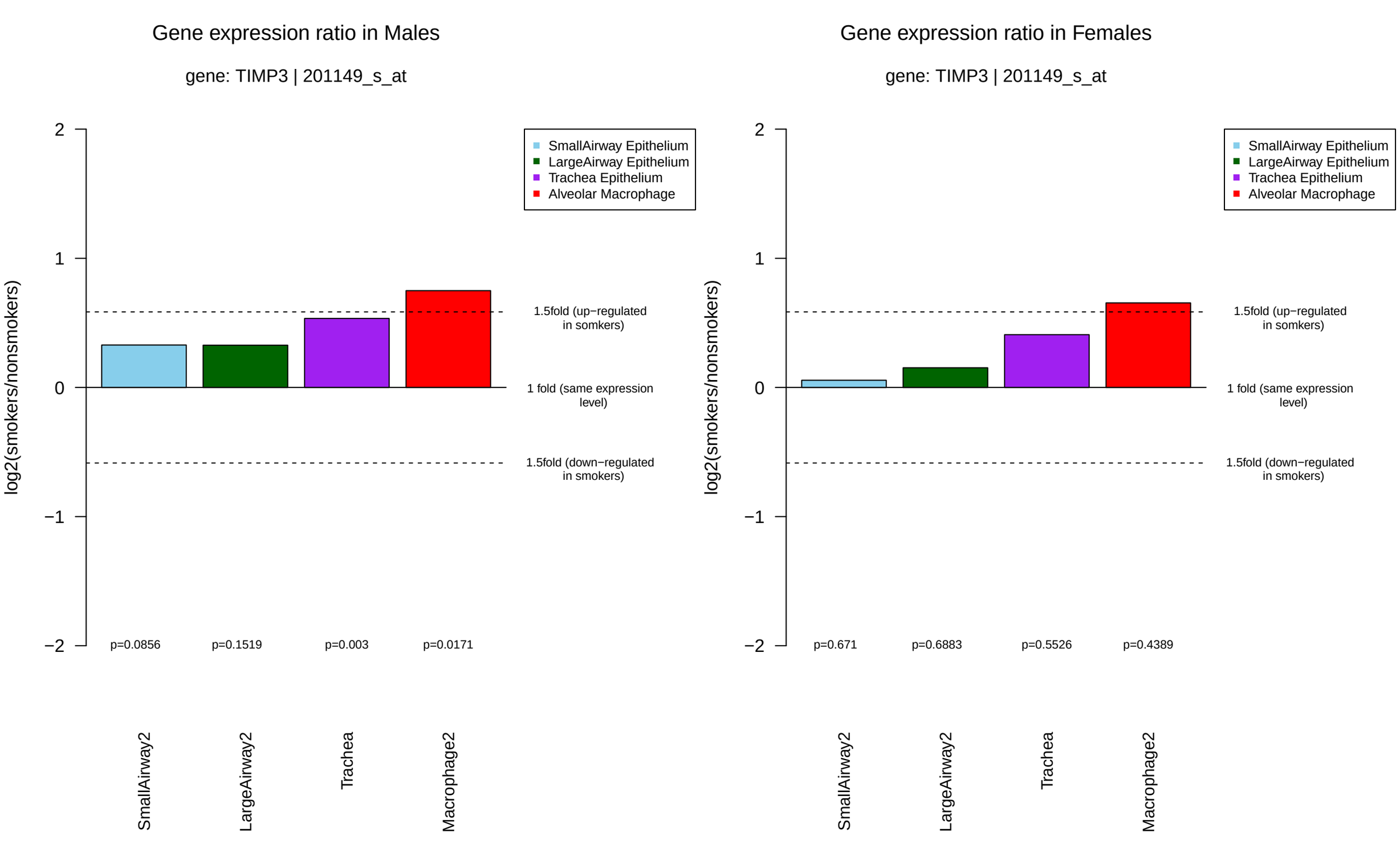
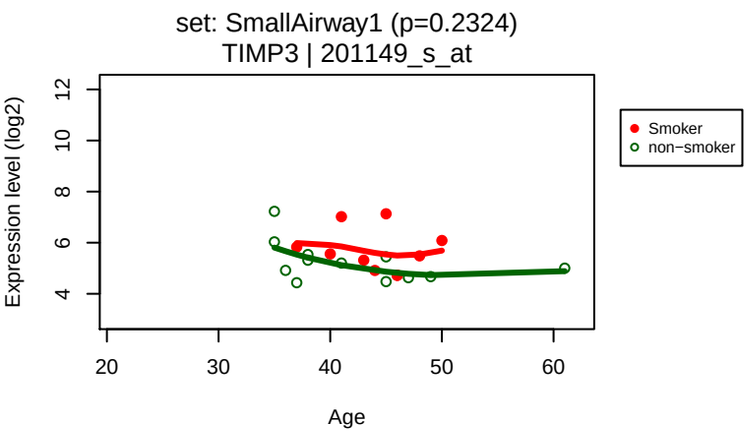
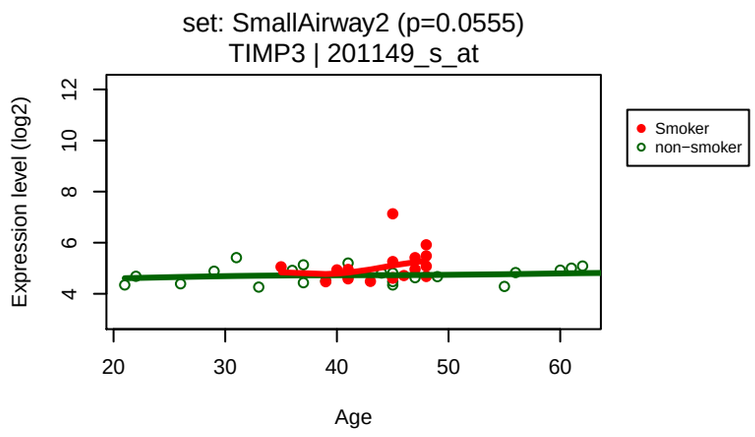
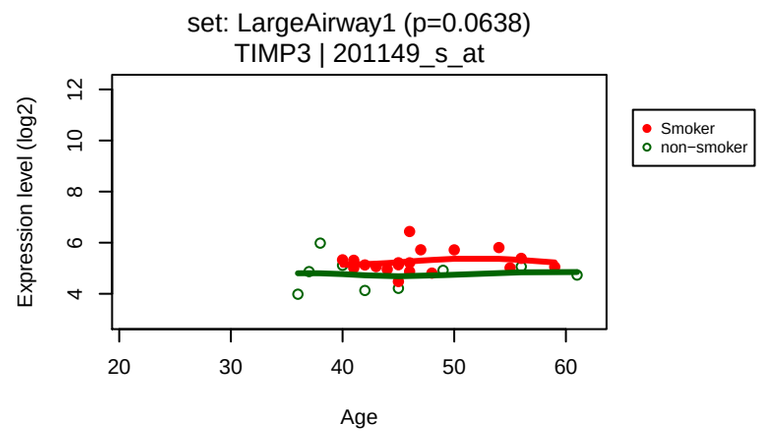

Gene expression of smokers and nonsmokers
gene: TIMP3 | 201149_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2324
p=0.0555
p=0.0116
p=0.0638
p=0.1313
p=0.0098
p=0.0254
p=0.03
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

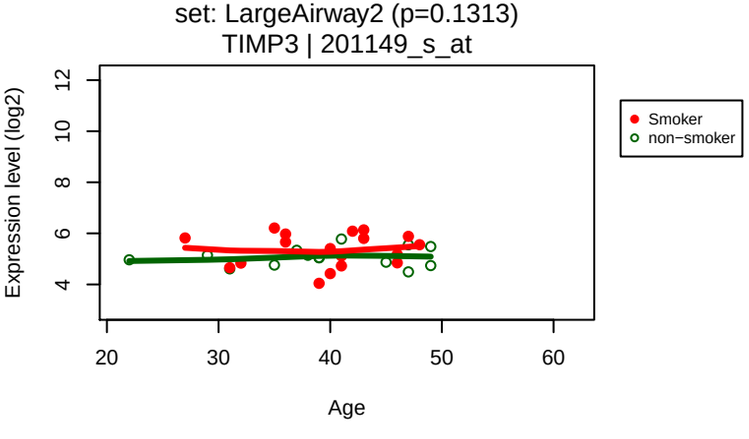

Gene expression ratio (smokers/nonsmokers)
gene: TIMP3 | 201149_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2324
p=0.0555
p=0.0116
p=0.0638
p=0.1313
p=0.0098
p=0.0254
p=0.03
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

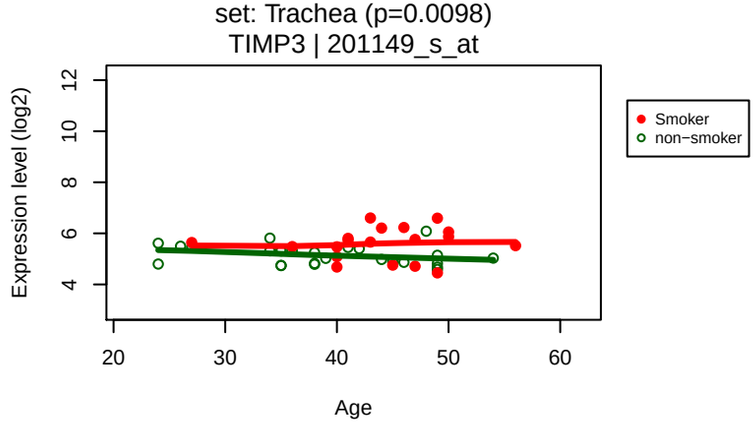

Gene expression ratio in Males
gene: TIMP3 | 201149_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0856
p=0.1519
p=0.003
p=0.0171
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIMP3 | 201149_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.671
p=0.6883
p=0.5526
p=0.4389
SmallAirway2
LargeAirway2
Trachea
Macrophage2

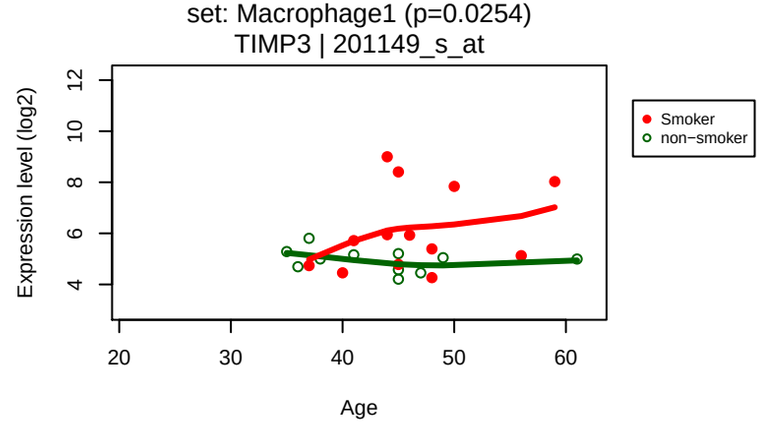

set: SmallAirway1 (p=0.2324)
TIMP3 | 201149_s_at
Expression level (log2)
Age
Smoker
non−smoker

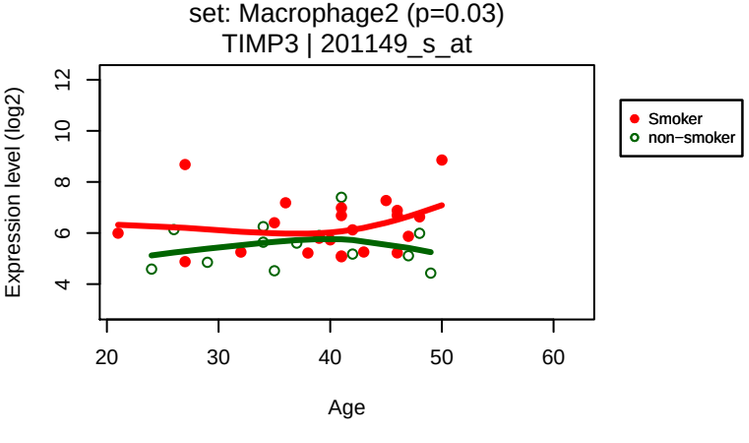

set: SmallAirway2 (p=0.0555)
TIMP3 | 201149_s_at
Expression level (log2)
Age
Smoker
non−smoker

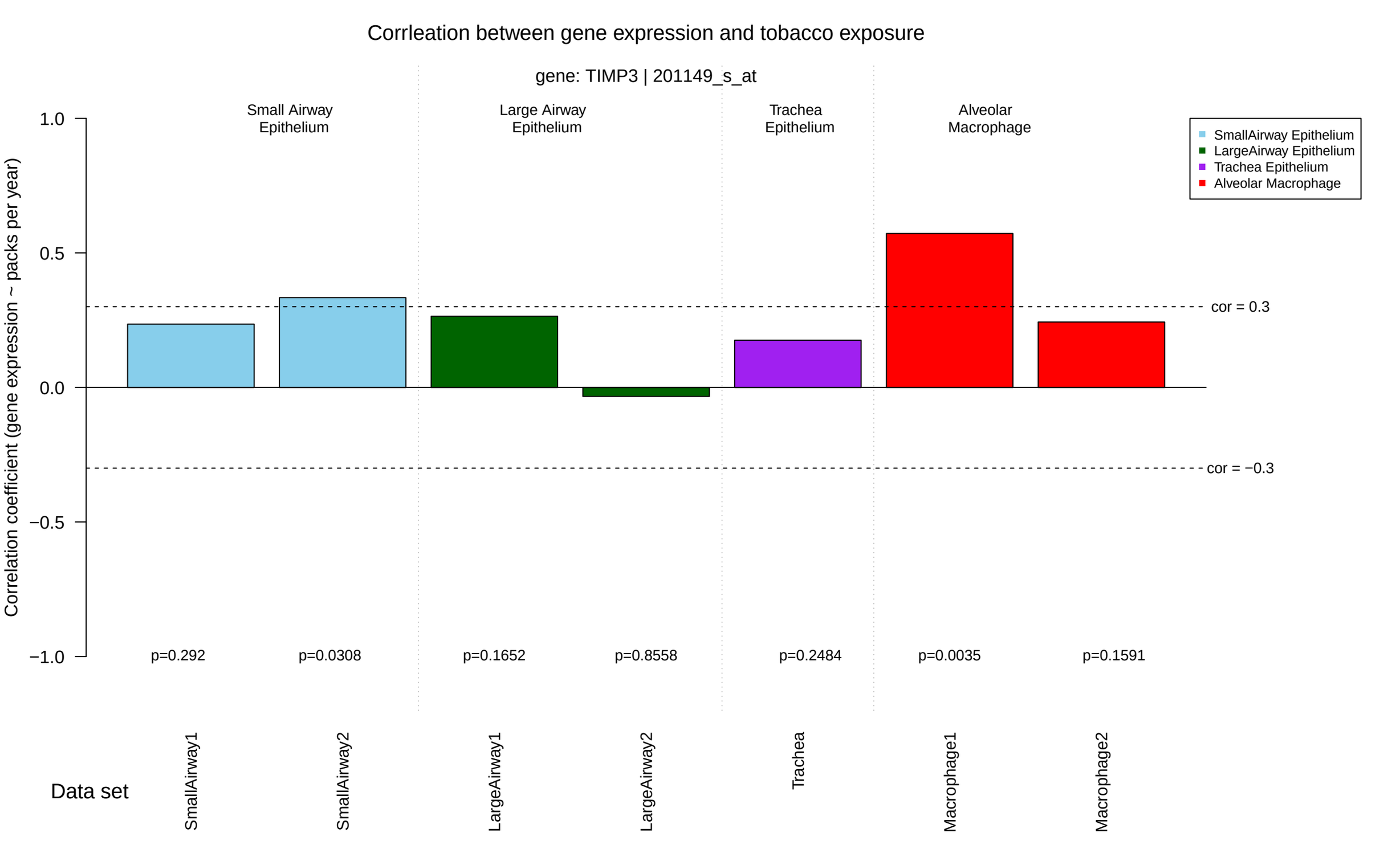

set: LargeAirway1 (p=0.0638)
TIMP3 | 201149_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1313)
TIMP3 | 201149_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0098)
TIMP3 | 201149_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0254)
TIMP3 | 201149_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.03)
TIMP3 | 201149_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIMP3 | 201149_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.292
p=0.0308
p=0.1652
p=0.8558
p=0.2484
p=0.0035
p=0.1591
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2